KAIZEN MATHS

Lesson starter

Transposition of Formulae: Changing the Subject

Powers and Roots

Name _____

S1

Use inverse operations carefully. Include plus or minus when square rooting a squared variable.

Make x the subject.

$y = 2x^2 + 2$

S2

Square both sides to remove the square root, then isolate the subject.

Make x the subject.

$y = \sqrt{x + 5}$

S3

Isolate the square root first, square both sides, then finish the rearrangement.

Make x the subject.

$y = 5\sqrt{x + 6} + 8$

S4

Work step by step through each formula involving a square root.

Make s the subject.

$v = \sqrt{u^2 + 2as}$

Show your working

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ALG-0852-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Powers and Roots

Name _____

EA1

Use inverse operations carefully. Include plus or minus when square rooting a squared variable.

Make x the subject.

$y = 3x^2 + 12$

EA2

Square both sides to remove the square root, then isolate the subject.

Make x the subject.

$y = \sqrt{x + 9}$

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Powers and Roots

Name _____

EB1

Isolate the square root first, square both sides, then finish the rearrangement.

Make x the subject.

$y = 4\sqrt{x + 4} + 9$

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Powers and Roots

Name _____

EA1

Use inverse operations carefully. Include plus or minus when square rooting a squared variable.

Make x the subject.

$y = 3x^2 + 12$

EA2

Square both sides to remove the square root, then isolate the subject.

Make x the subject.

$y = \sqrt{x + 9}$

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Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Powers and Roots

Name _____

EB1

Isolate the square root first, square both sides, then finish the rearrangement.

Make x the subject.

$y = 4\sqrt{x + 4} + 9$

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Show your working

Teacher answers & working

Transposition of Formulae: Changing the Subject · Powers and Roots

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S1

Use inverse operations carefully. Include plus or minus when square rooting a squared variable.

Make x the subject.

$y = 2x^2 + 2$

Answer:

$x = \pm\sqrt{\frac{y-2}{2}}$

Working:

Start with the formula. The subject is squared, so isolate x^2 first.

$y = 2x^2 + 2$

Subtract 2 from both sides.

$y - 2 = 2x^2$

Divide both sides by 2.

$\frac{y-2}{2} = x^2$

Take the square root of both sides. Because x^2 has two possible square roots, include ±.

$x = \pm\sqrt{\frac{y-2}{2}}$

S2

Square both sides to remove the square root, then isolate the subject.

Make x the subject.

$y = \sqrt{x+5}$

Answer:

$x = y^2 - 5$

Working:

Start with the formula. The subject is inside a square root.

$y = \sqrt{x+5}$

Square both sides to remove the square root.

$y^2 = x + 5$

Subtract 5 from both sides.

$y^2 - 5 = x$

Write x as the subject.

$x = y^2 - 5$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Powers and Roots

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S3

Isolate the square root first, square both sides, then finish the rearrangement.

Make x the subject.

$y = 5\sqrt{x+6} + 8$

Answer:

$x = \frac{y-8}{5}^2 - 6$

Working:

Start with the formula. Isolate the square root before squaring.

$y = 5\sqrt{x+6} + 8$

Subtract 8 from both sides.

$y - 8 = 5\sqrt{x+6}$

Divide both sides by 5.

$\frac{y-8}{5} = \sqrt{x+6}$

Square both sides to remove the square root.

$\frac{y-8}{5}^2 = x + 6$

Subtract 6 from both sides.

$x = \frac{y-8}{5}^2 - 6$

S4

Work step by step through each formula involving a square root.

Make s the subject.

$v = \sqrt{u^2 + 2as}$

Answer:

$s = \frac{v^2-u^2}{2a}$

Working:

Start with the formula and identify the subject s inside the square root.

$v = \sqrt{u^2 + 2as}$

Square both sides to remove the square root.

$v^2 = u^2 + 2as$

Subtract u^2 from both sides.

$v^2 - u^2 = 2as$

Divide both sides by 2a.

$s = \frac{v^2-u^2}{2a}$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Powers and Roots

ALG-0852-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

Use inverse operations carefully. Include plus or minus when square rooting a squared variable.

Make x the subject.

$y = 3x^2 + 12$

Answer:

$x = \pm\sqrt{\frac{y-12}{3}}$

Working:

Start with the formula. The subject is squared, so isolate x^2 first.

$y = 3x^2 + 12$

Subtract 12 from both sides.

$y - 12 = 3x^2$

Divide both sides by 3.

$\frac{y-12}{3} = x^2$

Take the square root of both sides. Because x^2 has two possible square roots, include ±.

$x = \pm\sqrt{\frac{y-12}{3}}$

EA2

Square both sides to remove the square root, then isolate the subject.

Make x the subject.

$y = \sqrt{x+9}$

Answer:

$x = y^2 - 9$

Working:

Start with the formula. The subject is inside a square root.

$y = \sqrt{x+9}$

Square both sides to remove the square root.

$y^2 = x + 9$

Subtract 9 from both sides.

$y^2 - 9 = x$

Write x as the subject.

$x = y^2 - 9$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Powers and Roots

ALG-0852-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Isolate the square root first, square both sides, then finish the rearrangement.

Make x the subject.

$y = 4\sqrt{x+4} + 9$

Answer:

$x = \frac{y-9}{4}^2 - 4$

Working:

Start with the formula. Isolate the square root before squaring.

$y = 4\sqrt{x+4} + 9$

Subtract 9 from both sides.

$y - 9 = 4\sqrt{x+4}$

Divide both sides by 4.

$\frac{y-9}{4} = \sqrt{x+4}$

Square both sides to remove the square root.

$\frac{y-9}{4}^2 = x + 4$

Subtract 4 from both sides.

$x = \frac{y-9}{4}^2 - 4$